



Rutile

△ Cabochon of rutile needles in quartz

Arguably, rutile is more important as a mineral that imparts desirable characteristics to other stones than it is as a gem in its own right. Rutile commonly forms microscopic, oriented inclusions in other minerals, and produces the asterism shown by star rubies and sapphires. It is also familiar as the golden, needle-like crystals trapped inside crystals of rutilated quartz, a material that has been used as an ornament since ancient times. Some reddish rutile crystals are darkly transparent and these are sometimes faceted for collectors.

Specification

Chemical name Titanium dioxide | **Formula** TiO_2 | **Colours** Reddish brown to red, golden | **Structure** Tetragonal | **Hardness** 6–6.5 | **SG** 4.2–4.3 | **RI** 2.62–2.90 | **Lustre** Sub-adamantine to sub-metallic | **Streak** Pale brown to yellowish | **Locations** Sweden, Italy, France, Austria, Brazil, USA

Reddish-brown rutile crystals



Quartz crystal with rutile | **Rough** | This example of natural, clear quartz crystal is shot through in all directions with multiple acicular (needle-like) crystals of rutile.

Multiple inclusions



Golden rutile | **Colour variety** | Rutile often takes on a bright golden colour when enclosed in quartz, as can be seen in this example with sheaf-like needles.

Rutile needles



Cabochon | **Cut** | This polished quartz cabochon shows a large number of inclusions of rutile needles forming a dense pattern beneath its surface.

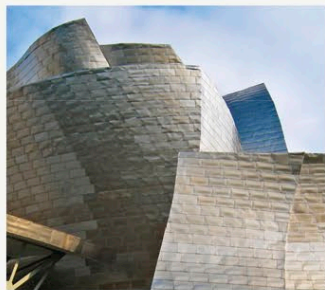


Scent bottle | **Carved** | This carved scent bottle is made from rutilated quartz, with fine golden rutile needles throughout. It is also adorned with gold and onyx.

Titanium from rutile

Everyday uses

Although it may not be a household name, rutile is a vital part of modern life. Titanium is derived from rutile and, because it does not react with organic tissues, it is used to make artificial joints for hip and knee replacements and other prosthetic devices. It is also extensively used in aircraft and other applications that require high strength, low density, and excellent corrosion resistance. Titanium dioxide, meanwhile, is the main white pigment in paint, plastics, and white enamel.



Guggenheim Museum Architect Frank Gehry used titanium as the surface for his curving building in Bilbao, Spain.

Reverse intaglio | **Carved** | Rutile needles are visible in the stopper of this scent bottle. The body is carved from crystal quartz with a reverse intaglio carving.

Rutile needles

Reverse intaglio carving of dolphin

Carved quartz body

